Team ByeByeHelloHaHaHa: A Generic, Render-Light Pipeline for 4D Human–Object Reconstruction from a Single Video

V2D Challenge 2026, Track 1 (Reconstruction) — Technical Report

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Abstract

I reconstruct, from one static third-person RGB video and nothing else, an MHR human sequence and a textureless object mesh with a per-frame similarity pose. Every clip goes through the same stages with the same settings: a vision-language model names the object, SAM 3 tracks the actor and the object, MoGe-2 and OpenD4RT give metric, temporally consistent depth, SAM 3D Body gives the human, and SAM 3D Objects gives the object mesh together with a pose-and-scale *layout* that I use as extra starting hypotheses for a render-free point fit. A contact-, floor- and penetration-aware joint placement and a final smoothing stage tuned to the acceleration metrics complete the pipeline. My reference run scores CD-H 14.24 cm, CD-O 26.93 cm, ACC-H 0.1375, ACC-O 0.0732 and PEN 0.0015 cm on the public leaderboards. I report the ablations that shaped it, including a render-and-compare variant that trades ACC-O for CD-O.

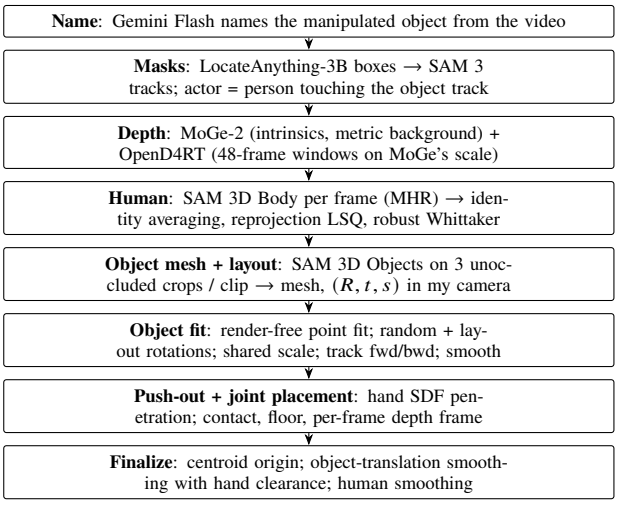


Figure 1: Track 1 pipeline (reference run `lay_jfin`). All stages run with fixed, object-independent settings.

1 Task and metrics

Track 1 provides 30 clips (10 objects, 3 clips each; 1536×1152, 30 fps, 16,563 frames of which 9,877 are scored) from a static camera, with no intrinsics, no object template and no depth. A submission is, per clip, MHR parameters (pose, scales, shape) for the actor and an object mesh (≤4,096 faces after packing) with a rotation, translation and scale per frame. The scorer aligns the prediction once, by a Sim(3) on the human at the first scored frame, then reports five metrics, each a separate leaderboard and each a mean over clips: Chamfer distance of the human (CD-H) and the object (CD-O), acceleration error of 22 human joints (ACC-H) and of the object translation (ACC-O), and the penetration depth of 512 hand points into the object (PEN). The final ranking sums the five ranks.

Two properties of the metrics guided my design. (i) Because alignment happens once, any drift in scale or depth after the first scored frame is never forgiven. (ii) ACC-O uses only the object translation, i.e. the path of the mesh origin, and a zero-acceleration prediction scores exactly the mean ground-truth acceleration (≈0.074 on the leaderboard). A noisy track is therefore worse than a static one, and the mesh origin should be the object's centre.

2 Pipeline

Figure 1 shows the stages. A single driver (`scripts/e2e_vm.py`) runs them on one Colab GPU VM; stages resume from their outputs, and nothing in them depends on which object a clip shows.

2.1 Object name and masks

Gemini Flash is asked which object the person interacts with; the short answer is the only object-specific input. LocateAnything-3B grounds the name on ∼11 keyframes, and the boxes seed SAM 3 video tracks, which the boxes then vote on (moving tracks win over still ones, so a table the hand rests on is not chosen). The actor is the foreground person whose mask touches the object track, in the image and in depth, on most scored frames. If no track passes a coverage and motion check, a motion-seeded or depth-foreground fallback is used.

2.2 Depth and camera

MoGe-2 runs per frame; one field of view per clip (the median of free estimates on 16 frames) gives the intrinsics, and the

static background, a median over the clip, gives a metric reference. OpenD4RT produces temporally consistent depth in 48-frame windows (overlap 16); each window is scaled to the MoGe background on static pixels, windows are blended, and each frame is stabilized against the background. The camera is static, which gives a check on OpenD4RT: its camera extrinsics should be the identity, and they nearly are. A Sim(3) fitted on background points from each frame to its window's first frame has a median rotation of 0.12° and a median translation of 0.2% of the scene depth over the 30 clips (worst 0.63° and 1.2%), which supports OpenD4RT's geometry as a reliable depth source. A floor plane is fitted for the joint stage.

2.3 Human

SAM 3D Body runs per frame on the actor box and mask with my intrinsics and returns native MHR parameters. Post-processing fixes one identity (mean scales and shape over fully visible frames), re-solves each translation so the 22 scored joints reproject where SAM 3D Body put them, and applies a robust Whittaker smoother (second differences) to translation, root rotation and pose, down-weighting and re-solving outlier frames. I keep SAM 3D Body's own depth for the body: moving it along the ray to the depth map cost 2 cm CD-H (Table 2). The body then defines the metric frame: the ratio of depth-map to body depth over the actor mask rescales the depth used for the object.

2.4 Object mesh and layout

Three crops per clip are taken from the best unoccluded frames (large mask, little outline shared with the actor, ≥30 frames apart). For each, SAM 3D Objects receives the full frame, the full-resolution mask and a pointmap from my depth at the body's scale, and returns a mesh and a layout $x_{\text{cam}} = sRv + t$ in my camera. Meshes are cleaned (floaters removed, main part kept), normalized to unit size and decimated to ≤4,000 faces. The layout is re-expressed for the normalized mesh.

2.5 Object pose fit

The pose model is $x_{\text{cam}} = sRp + t$ for points p sampled on the mesh; no rasterizer is used. Four robust (Geman–McClure) terms compare it with the observations: model points falling outside the object ∪ actor mask (occlusion by the actor is free), object-mask pixels not covered by projected points, interior object depth to the nearest model point, and front-facing model points against the depth map. Per clip I (1) screen each candidate mesh from many rotations on a few well-visible keyframes, adding each SAM 3D layout from that mesh as a start (and its 25° perturbations) on its own frame; (2) fix one scale, the weighted median of keyframe scales; (3) track forward and backward from the best keyframe, each frame starting from the previous pose and six rotations of it, with a rotation-jump penalty; and (4) refine all frames jointly with second-difference penalties on t and R.

run	CD-H	CD-O	ACC-H	ACC-O	PEN
`lay_jfin` (mine)	14.24	26.93	0.1375	0.0732	0.0015

Table 1: Public leaderboard scores of the reference run (cm; lower is better).

2.6 Penetration push-out and joint placement

The packed mesh is turned into a signed-distance grid with the scorer's inside rule (generalized winding numbers). A refit adds a hinge on the signed distance of every hand vertex the MHR finger parameters move, keeping the data terms so the object stays where the video puts it. The joint stage then places the object relative to the body: a per-frame depth ratio between the depth map and the rendered body (smoothed) defines one consistent depth frame; contact candidates are hand and sole points projecting onto the object mask; a search along the camera ray ($s \to as$, $t \to at$, which leaves the image unchanged) trades depth agreement, contact, floor and a scale prior; a ray-only per-frame refinement follows, and discrete rotation hypotheses resolve poses the image cannot separate (e.g. a ring's two tilts).

2.7 Finalization

The mesh origin is moved to the surface centroid (the pose compensates, so the surface is unchanged), so that ACC-O follows the object's centre. The object translation is then re-solved as $\arg\min_t \|t - t_0\|^2 + \lambda_o \|D_2 t\|^2 + w \sum \text{relu}(m - \text{SDF}(\text{hand}))^2$ with $\lambda_o = 10^5$, rotation and scale kept, and the human receives one more Whittaker pass ($\lambda_{t,r,p} = 1000, 600, 600$).

3 Results

Table 1 gives the reference run. Each leaderboard ranks my best submission on it separately; the reference run is the single pipeline I describe and reproduce.

Ablations. Table 2 lists leaderboard results of the variants that led to the reference run. Rows marked with a dash were submitted to a subset of boards only.

What mattered. *Object meshes* were the largest CD-O lever: replacing Hunyuan3D-2.1 by SAM 3D Objects moved CD-O from 33.3 to 28.6 cm without any other change (A vs. H). *Acceleration-aware finalization* took ACC-O from 0.119 to 0.073–0.075, i.e. to the level of a static object, for no measurable CD-O cost (A–C); coupling the object to the wrists hurt (D vs. C), since per-frame wrist estimates are noisier than the object's true motion. *Human smoothing* brought ACC-H from 0.170 to 0.137 for +0.04–0.07 cm CD-H (E, I–M); moving the body to the depth map hurt CD-H by 2 cm (F), so the

	variant	CD-H	CD-O	ACC-H	ACC-O	PEN
A	generic pipeline, Hunyuan3D-2.1 meshes, no finalization (`e2e_box`)	14.20	33.34	0.170	0.119	0.00091
B	A + finalization, $\lambda_o = 10^3$, no human smoothing	14.20	33.25	0.170	0.0876	0.00089
C	A + finalization, $\lambda_o = 10^4$	–	–	–	0.0749	–
D	C + object follows the holding wrists' acceleration	–	–	–	0.0848	–
E	B + human Whittaker $(10^3, 200, 200)$	14.27	–	0.1374	–	–
F	body depth moved along the ray to the D4RT actor depth	16.16	–	0.172	–	–
G	A + render-and-compare refinement + finalization	–	30.30	–	–	–
H	SAM 3D web-demo meshes (RGB only), no layout seeds	–	28.63	–	–	–
I	H + render-and-compare + finalization (`s3dall_rf_fin`)	14.24	**24.48**	0.1375	0.0748	0.00126
J	SAM 3D on my crops + layout seeds (`lay_jfin`, reference)	14.24	26.93	0.1375	**0.0732**	0.00148
K	J + render-and-compare	14.24	26.63	0.1375	0.0749	**0.00084**
L	web-demo meshes + layout seeds transferred by ICP	14.24	28.52	0.1375	0.0737	0.00264
M	L + render-and-compare	14.24	26.37	0.1375	0.0746	0.00105

Table 2: Leaderboard ablations (cm). A–G use Hunyuan3D-2.1 meshes; H–M use SAM 3D Objects meshes. Final runs (I–M) use $\lambda_o = 10^5$ and human smoothing $(1000, 600, 600)$.

body, not the depth map, defines the metric frame. *Render-and-compare refinement* (nvdiffrast, masks only, with an occlusion rule against the rendered body) helps CD-O on top of a render-free fit (G: −3 cm; I vs. H: −4.1 cm) but slightly roughens the object track (ACC-O +0.002, J vs. K), which is why the reference run omits it. *Layout seeds* gave the best ACC-O (J) but not the best CD-O: with the same web-demo meshes, adding the seeds cost ≈2 cm CD-O (I vs. M), while the mesh source mattered little once seeds were used (K vs. M).

Reference run choice. Variant I has the best CD-O and J the best ACC-O; their summed ranks over the three object boards are within two places. I report J because it is fully automatic: variant I relies on meshes generated interactively in the SAM 3D web demo, which a single-command reproduction cannot rerun. In the web demo I selected the object so that its mask matched my SAM 3 mask, so in principle running SAM 3D Objects locally on the same frame and mask, from RGB only as the demo does, should reproduce those meshes and make variant I automatic as well; I had no compute units left to verify this.

4 Failure cases and limitations

I picked the failures from per-clip proxy metrics of the reference run (object mask IoU and depth error against my own masks and depth, since the leaderboard reports only means). Three clips stand out; Figure 2 shows each in the input view and from a side camera.

Thin ring without depth (clip 1, hula hoop; object IoU 0.19). The depth map does not resolve a 5 cm tube, so only 7 of 341 frames give object depth and the image fixes the hoop only up to a scale along the ray. The joint stage then sets that scale from hand contact alone and shrinks the hoop by 24%, from the 0.90 m of the image fit to 0.69 m (the image fits of the other two hoop clips give 0.84 m), which pulls it toward the camera: the leg goes through the hoop in the video but passes beside it in the reconstruction.

Object occluded in every crop (clip 2, hula hoop; IoU 0.41). The hoop is spun around the body, so all three crops show a ring cut by the torso. SAM 3D Objects reconstructs from these partial rings and its layouts fall largely outside the mask (35–43% of their points inside, against 74–100% for the other two hoop clips); the fit loss is ten times that of clip 0. The resulting track keeps the ring's outline in the image but places it upright through the torso, which no term rejects, since penetration is only checked for hand points.

Flat object seen edge-on (clip 14, tablet; IoU 0.42). The tablet lies on a table nearly in line with the camera, so its mask is a thin sliver. The three SAM 3D layouts disagree on its size by a factor of 2.5 (0.18–0.45 m), the joint stage switches between rotation hypotheses, and in the unheld frames the tablet sits near the floor in front of the table instead of on it.

More generally, the fit has no appearance term, so symmetric or thin objects can keep a wrong rotation that the masks cannot reject; the joint stage's discrete hypotheses only cover known ambiguities. Object scale comes from a single depth frame tied to the body, so a biased body depth biases the object. A noisy object track is clipped toward static by the finalization, which helps ACC-O but cannot recover true fast motion.

I see depth resolution as the main limit of the current approach. OpenD4RT, the open reimplementation of D4RT I use, takes 256×256 inputs, i.e. the 1536×1152 frame downsampled about 6× horizontally, so thin and small objects cover only a few depth samples: the hoop's tube in clip 1 is one example, and the edge-on tablet in clip 14 another. I did not have access to the original D4RT weights; with them I would expect better object depth on such objects, and with it better object placement and CD-O.

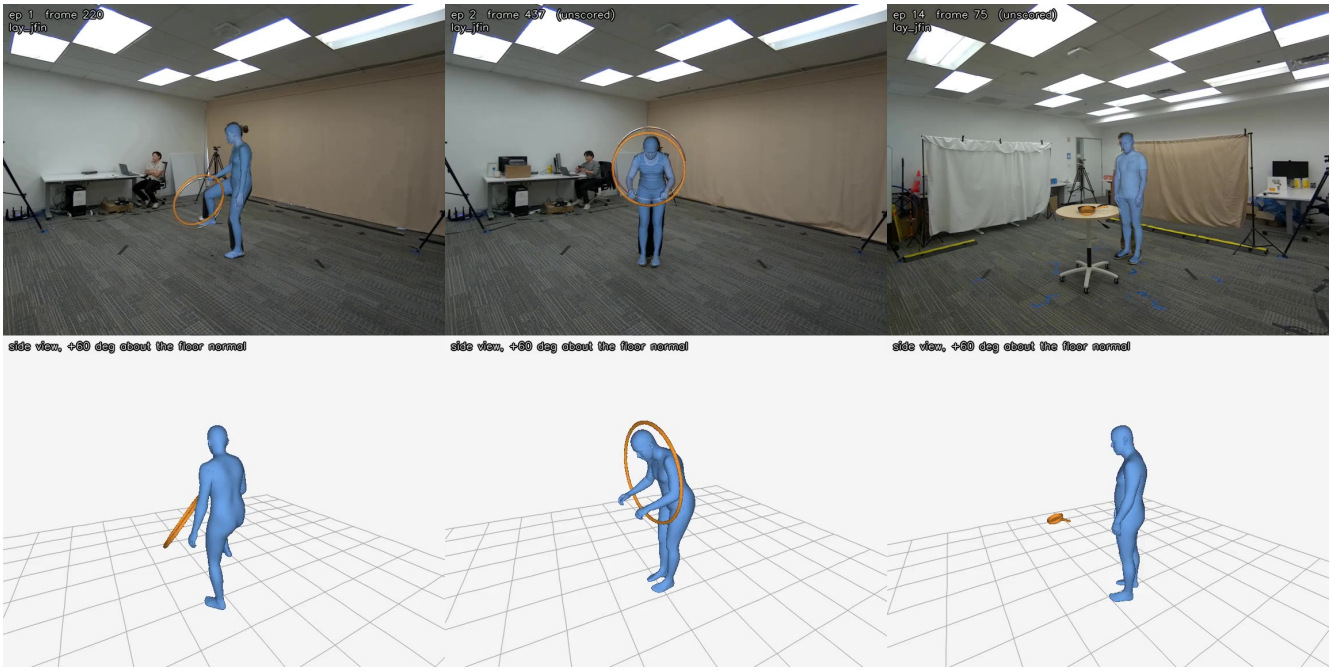


Figure 2: Failure cases of the reference run (human blue, object orange). Top: input frame with the reconstruction; bottom: side view, 60° about the floor normal. Left: clip 1, the hoop, shrunk by 24% in the joint stage, hangs beside the leg that goes through it in the video. Middle: clip 2, the ring is placed upright through the torso. Right: clip 14, the tablet is placed near the floor instead of on the table.

5 Compute and reproduction

All model inference ran on Google Colab (L4 at 1.54 and A100 at 5.3 compute units per hour); the laptop (6 GB GPU) ran only orchestration, light geometry and the render-and-compare variants. SAM 3D Objects takes 13–40 s per crop on an A100 (90 crops); the object to joint stages for 30 clips take about 3 h on one L4, and finalization a few minutes.

Code: `https://github.com/ChanHaoHao/4DHumanObjectInteraction`, commit `ea90fbc` (branch `e2e-box`). On a Colab A100 session (`colab new -s e2e --gpu A100`), one command runs every stage for all 30 clips and pulls the results to `results/lay/episodes`: `python3 scripts/e2e_colab.py run e2e --run lay --episodes 0-29 --sam3d`. External models: Gemini Flash, LocateAnything-3B, SAM 3, MoGe-2, OpenD4RT, SAM 3D Body, SAM 3D Objects; no training.